

PD Group:

Name:

**ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS**

**BIOLOGY
Higher 2**

9648/01

21 SEPT 2011 WEDNESDAY

1 hour 15 minutes

READ THESE INSTRUCTIONS FIRST

Paper 1 Multiple Choice

Write and/or shade your name, NRIC / FIN number and PD group on the MCQ OAS in the spaces provided.

Answer **all** questions.

For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

At the end of the examination, only hand in the MCQ OAS

There are forty questions in this section. Answer **all** questions.
For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice.

- 1 A certain organelle in a eukaryotic cell was isolated and analyzed. It was found that the organelle contains proteins, nucleotides and phospholipids.

Which of the following organelles can it be?

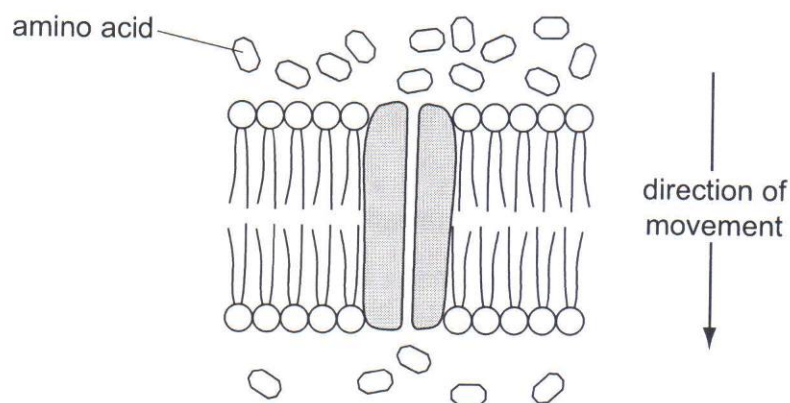
- I. Nucleus
- II. Lysosome
- III. Ribosome
- IV. Rough endoplasmic reticulum

- A I only.
- B III only.
- C IV only.
- D I and IV only.

- 2 Which of the following is true about organelles?

- A Vesicles from the smooth endoplasmic reticulum carry lipids to the Golgi apparatus where they are modified.
- B Mitochondria and chloroplasts can be seen under a light microscope.
- C Mitochondria share the same 80s ribosome as the prokaryotic cell.
- D Hydrolytic enzymes are found in lysosomes in animal and plant cells.

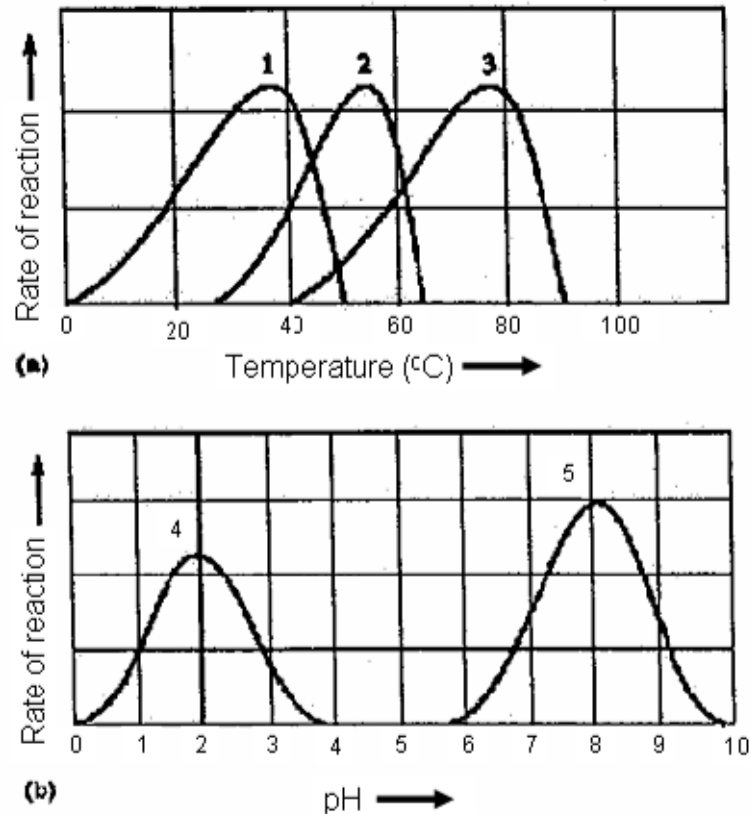
- 3 The diagram represents part of a cell surface membrane through which amino acid molecules are moving.



Which correctly best explains this movement?

A	simple diffusion	amino acids are small in size
B	simple diffusion	R groups of some amino acids are hydrophobic
C	facilitated diffusion	amino acids are polar molecules
D	facilitated diffusion	amino acids are large in size

- 4 The following graphs show the activities of different enzymes (1 to 5) under different conditions.



Which statement is a correct explanation for the rate of reaction of different enzymes?

- A Kinetic energy of enzyme 3 and its substrate is highest at 75°C.
- B At pH 2, most of the R groups at the active site of enzyme 4 are all negatively charged.
- C At pH 8.1, substrate is bonded to the active site of enzyme 5 by hydrogen bonds only.
- D At 60°C, several hydrogen bonds between R groups of enzyme 2 are broken.

- 5 Which of the following correctly describe(s) enzymatic inhibitions?

- I. All non-competitive inhibitors are allosteric inhibitors.
- II. Non-competitive inhibitors can bind reversibly or irreversibly.
- III. Feedback inhibition can only be found in metabolic pathway which involves at least one intermediate.
- IV. The reversible nature of competitive inhibition explains why a competitive inhibition effect can be overcome by high substrate concentration.

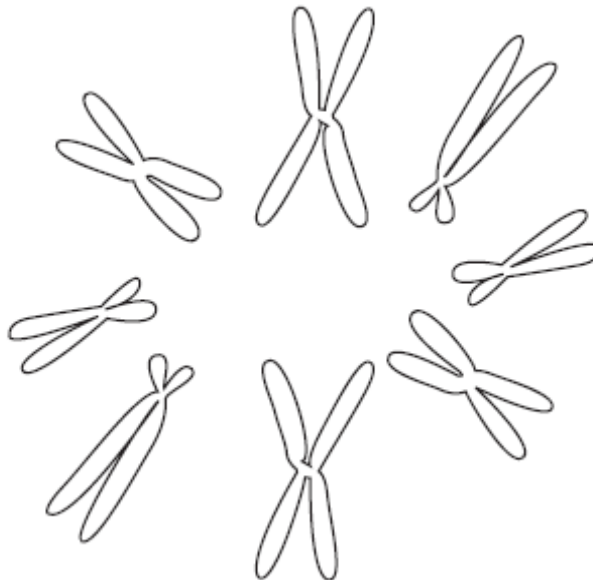
- A I and II
- B I and III
- C II, III and IV
- D All of the above.

6 Which of the following illustrates how ATP is being used?

	Process	How ATP is being used
I	Glycolysis	Hexose phosphorylation
II	Glycolysis	Substrate-level phosphorylation
III	DNA replication	Unzipping of double-stranded DNA by helicase
IV	Protein synthesis	Translocation of ribosome down a codon along mRNA

- A I and III
- B II and IV
- C III and IV
- D All of the above

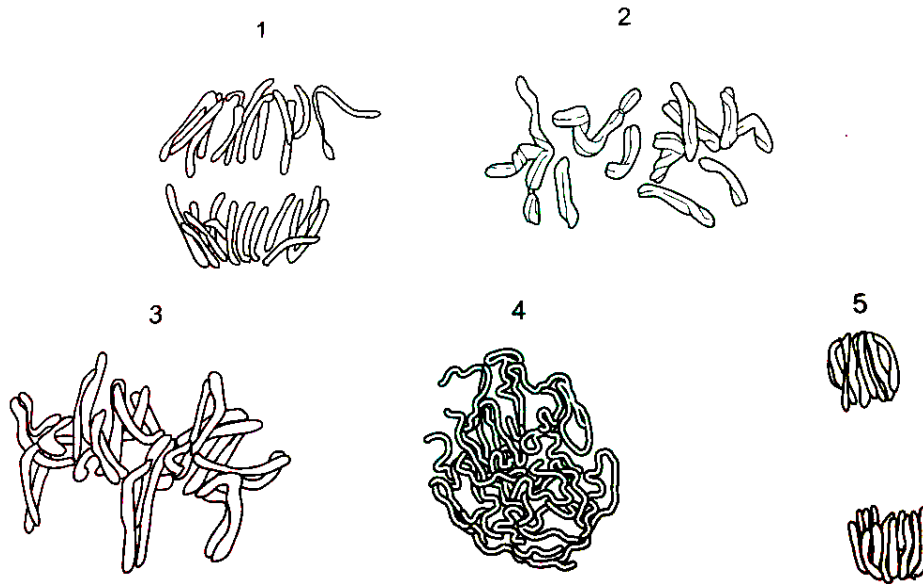
7 The diagram shows chromosomes at metaphase of mitosis.



What are the diploid and haploid numbers for this species?

	diploid	haploid
A	4	8
B	8	4
C	8	16
D	16	8

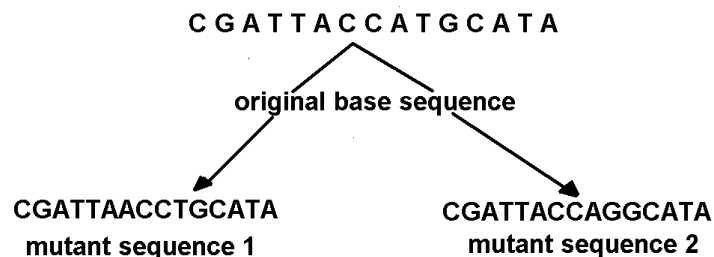
- 8 The drawing show stages of the mitotic cell cycle.



What would be correct?

	Stage of mitosis	Name of the stages	Description
A	1	Anaphase	Non-kinetochore microtubule lengthens, pulling chromosomes to opposite poles.
B	2	Prophase	The chromosomes move towards the equator to line up end to end of each other.
C	3	Metaphase	The homologous chromosomes pairs up at the equator.
D	4	Prophase	DNA replicates, each will form 2 DNA daughter molecules which condensed to form 2 identical sister chromatids.

- 9 The diagrams below show the results of two types of gene mutation.



Which one of the following shows the types of gene mutation producing mutant sequences 1 and 2?

	1	2
A	Addition	Deletion
B	Inversion	Substitution
C	Addition	Substitution
D	Substitution	Inversion

- 10** Which of the following statements about the newly synthesized strand of a human chromatid is correct?
- A** It was synthesized from multiple origins solely on continuous DNA synthesis.
 - B** It was synthesized from multiple origins solely on discontinuous DNA synthesis.
 - C** It was synthesized from multiple origins by a mixture of continuous and discontinuous DNA synthesis.
 - D** It was synthesized from multiple origins by either continuous or discontinuous DNA synthesis, depending on which specific sister chromatid is being examined.
- 11** In *E. coli* bacteria, regulation of gene expression involving the catabolite activator protein (CAP), is actually a type of positive regulation because
- A** cAMP-CAP helps RNA polymerase to initiate transcription.
 - B** CAP increases the production of cAMP.
 - C** glucose binds to CAP and activates it.
 - D** glucose stimulates the production of cAMP.
- 12** The following coding sequence is taken randomly from a bacterial genome.

3' – TTACGCTTCGAAATAGGAATATCATAGGCT-5'

Arg	CGU, CGC, CGA, CGG, AGA, AGG	Leu	UUA, UUG, CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Start	AUG	Ser	UCU, UCC, UCA, UCG, AGU, AGC
Stop	UAG, UGA, UAA	Tyr	UAU, UAC

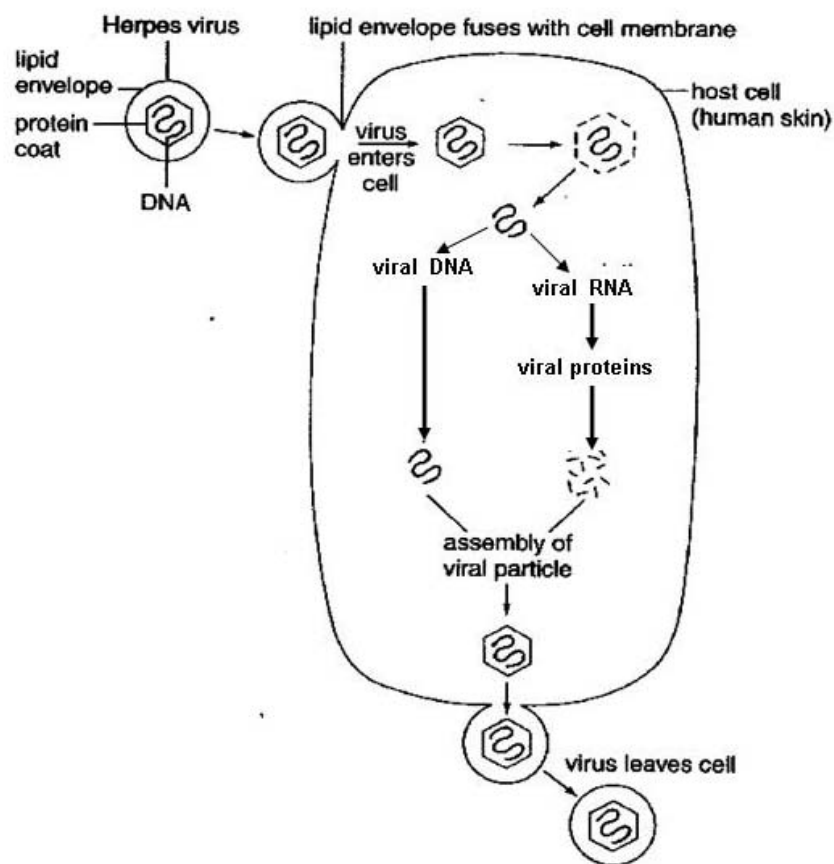
This sequence is cloned into a plasmid and transformed into a suitable host. What would be the first four amino acids of a peptide generated from this sequence as expressed by the host?

- A** Met-Arg-Ser-Phe
- B** Met-Arg-Ser-Lys
- C** Met-Ile-Phe-Leu
- D** Met-Tyr-Lys-Asp

- 13 The tobacco mosaic virus has RNA rather than DNA as its genetic material. In a hypothetical situation, RNA from a tobacco mosaic virus is mixed with proteins from a related DNA virus Z, resulting in a hybrid virus. If this hybrid virus were to infect a cell and reproduce, what would the resulting 'offspring' viruses be like?

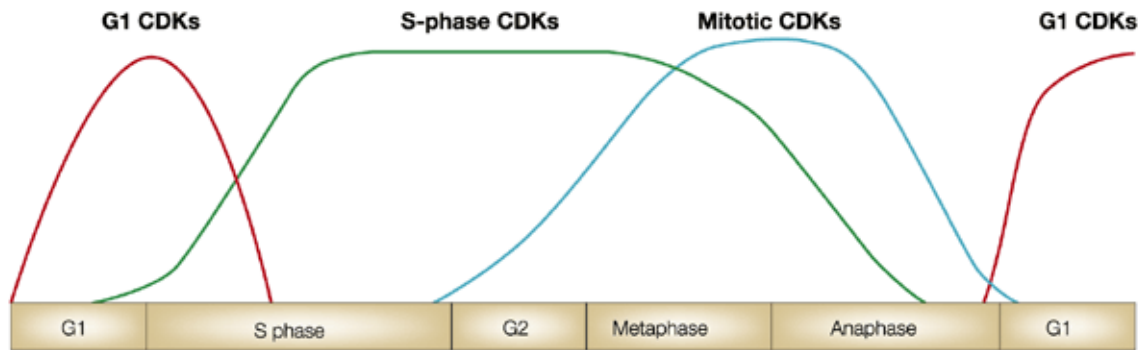
A Tobacco mosaic virus
 B Virus Z
 C A hybrid virus with tobacco mosaic virus RNA and virus Z protein
 D A hybrid virus with virus Z DNA and tobacco mosaic virus protein

- 14 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth. With reference to the diagram below, which of the following statements best describes the herpes virus?



A It is not a retrovirus as it does not contain RNA as its genetic material.
 B Its mode of replication is similar to that of HIV.
 C Its replication cycle includes a lysogenic phase.
 D Death of the host cell is necessary for the release of the viral progeny.

- 15** Cyclins are regulatory proteins that associate with cyclin-dependent kinases (CDKs) to control the different stages of the cell cycle. The right type and the correct amount of cyclins and CDKs must be present at the different stages to ensure regulation of the cell cycle. The concentrations of the different CDKs are shown in the diagram.



How could the levels of the different CDKs be regulated?

- I. Formation of heterochromatin.
- II. Binding of repressor to operator.
- III. Length of mRNA poly(A) tail.
- IV. Ubiquitylation of CDKs.

- A I and IV.
- B II and III.
- C III and IV.
- D I and II.

- 16** DNase is added to samples taken from the same chromosomal region from three different cell types. After allowing for digestion of DNA, the remaining DNA samples are weighed with the following results.

Cell type	Mass of DNA before adding DNase/ μg	Mass of DNA after adding DNase/ μg
A	5.0	5.0
B	5.0	3.4
C	5.0	4.1

What do these results suggest about genes and their expressions in all the three different cell types?

- A Some regions of the cell type A chromosomes contain RNA.
- B The genes found in the same chromosomal region in all the three cell types are expressed the most in cell type B.
- C The mass of DNA in cell type A did not decrease because DNA is not of complementary shape to the active site of DNase.
- D There are more phosphodiester bonds in the DNA found in the chromosomes of cell type B than those in cell type A and C.

- 17** Some viruses cause cancer in a few steps. An example of such a virus is a retrovirus known as human T-cell leukemia virus (HTLV). The following steps occur in a host cell resulting in cancer.

- 1** integration of viral control element into host DNA.
- 2** viral reverse transcriptase synthesizes a double-stranded DNA.
- 3** host growth factor proteins increase in amount that is more than usual.
- 4** host tumour suppressor gene products fail to regulate cell cycle.

Which is the correct order of steps to explain how HTLV causes cancer?

- A** 4 → 2 → 1 → 3
- B** 4 → 3 → 2 → 1
- C** 1 → 2 → 3 → 4
- D** 1 → 2 → 4 → 3

- 18** In *Drosophila*, vestigial wings and ebony body colour are recessive to the wild type and are not linked. Crossing two flies heterozygous for both characters produced 256 offspring.

How many offspring would be expected to have normal wings with ebony body colour?

- A** 144
- B** 96
- C** 48
- D** 16

- 19** In a certain type of plants, a recessive mutant allele leads to oval leaves as opposed to the normal round leaves. A second recessive mutant allele at a separate locus leads to dwarf plants as opposed to normal tall plants.

A cross between two round-leafed, tall plants which are heterozygous at both gene loci is carried out.

What is the probability of producing true-breeding plants with respect to the two characters?

- A** 12.5%
- B** 25 %
- C** 50%
- D** 75%

- 20** Chromosomal mutations were induced to produce a fertile hybrid species from cabbage and radish.

The table below shows the chromosome numbers in the parental species and the hybrids.

type of cell	number of chromosomes per cell
parental cabbage	18
parental radish	18
parental gametes	9
F ₁ hybrids	18
F ₁ gametes	9
F ₂ hybrids	18
F ₂ gametes	18
F ₃ hybrids	36

At which stage did the chromosomal mutation occur?

- A** during the formation of the F₁ gametes
 - B** during the formation of the F₂ gametes
 - C** during the fusion of the parental gametes
 - D** during the fusion of the F₁ gametes
- 21** Aneuploidy is the condition where the individual's genome has an extra chromosome or a missing chromosome.

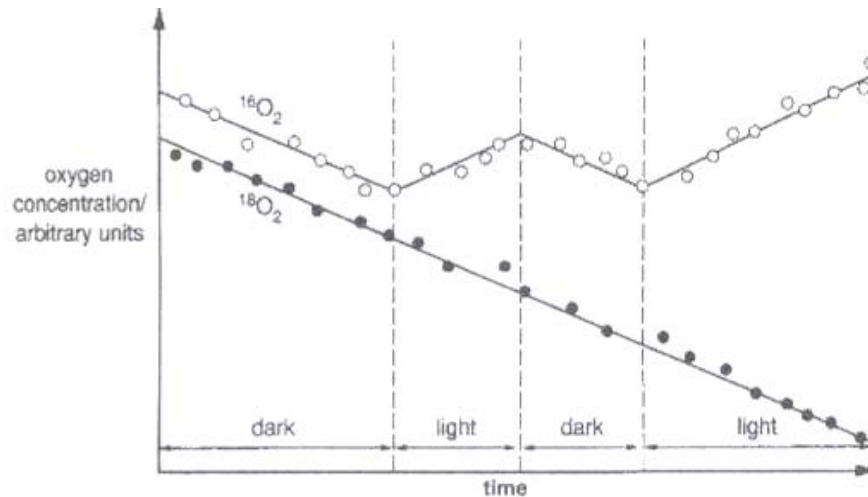
The sex chromosome combination XYY is found in a small proportion of men who are aneuploid. Such a combination is possible if

- A** one contributory gamete to the zygote is a sperm produced by non-disjunction at meiosis I.
- B** one contributory gamete to the zygote is a sperm produced by non-disjunction at meiosis II.
- C** one contributory gamete to the zygote is an egg produced by non-disjunction at meiosis II.
- D** two sperms fused with an egg.

- 22** Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension was measured in light and dark conditions.

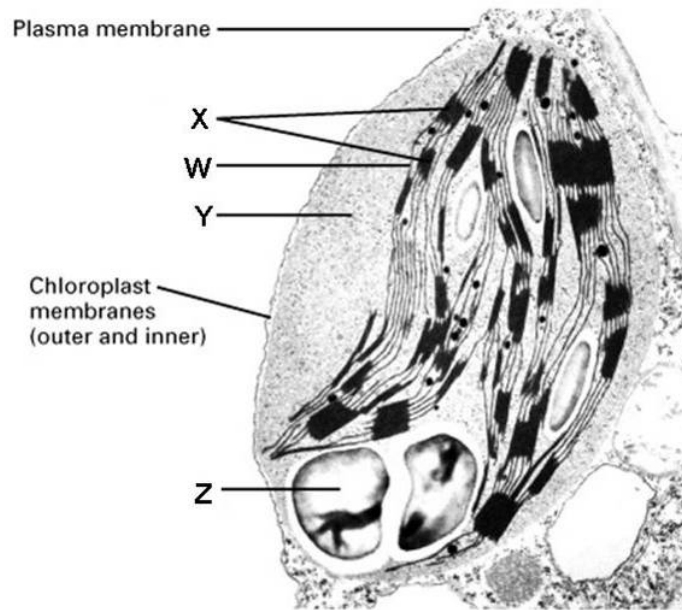
The graph shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A** H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
- B** $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.
- C** $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
- D** $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.

- 23 The electron micrograph shows the ultrastructure of a chloroplast as seen in section.



What are the functions of X, Y and Z?

	X	Y	Z
A	carbohydrate synthesis	Light absorption	Carbohydrate storage
B	Light absorption	carbohydrate synthesis	Carbohydrate storage
C	Light absorption	Carbohydrate storage	carbohydrate synthesis
D	Carbohydrate storage	carbohydrate synthesis	Light absorption

- 24 During aerobic respiration, ATP can be formed by oxidative phosphorylation in the electron transport system and by glycolysis.

In the complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

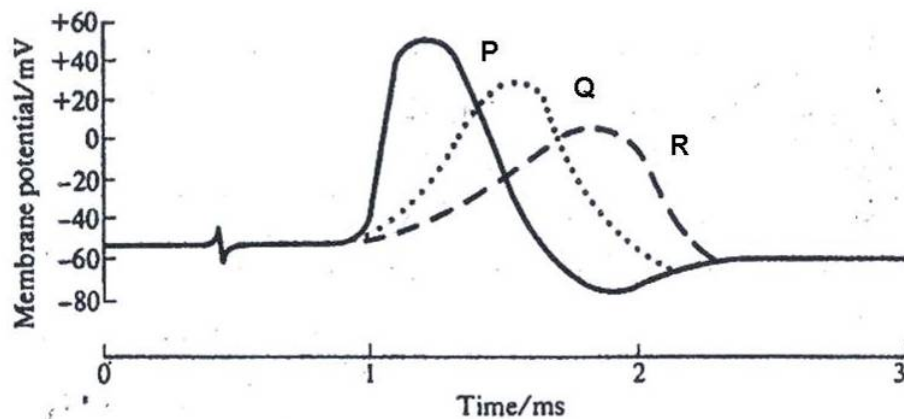
- A 90%
- B 70%
- C 30%
- D 15%

- 25 An inhibitor of the enzyme catalyzing the oxidative decarboxylation of α -Ketoglutarate (5C) is added to respiring animal tissues.

What will be the effect on the concentration of oxaloacetate, pyruvate and ATP?

	<i>oxaloacetate</i>	<i>pyruvate</i>	<i>ATP</i>
A	decreases	increases	decreases
B	decreases	decreases	increases
C	increases	increases	decreases
D	increases	decreases	increases

- 26 In 1949, Hodgkin and Katz investigated the effect of Na^+ ions on the production of action potentials in squid axons. Intracellular microelectrodes recorded action potentials from axons bathed in different concentrations of isotonic sea water. The results are shown in the figure below.



Which action potentials correspond with axons placed in normal sea water, one-half sea water and one-third sea water?

	P	Q	R
A	one-third	one-half	normal
B	one-half	normal	one-third
C	normal	one-third	one-half
D	normal	one-half	one-third

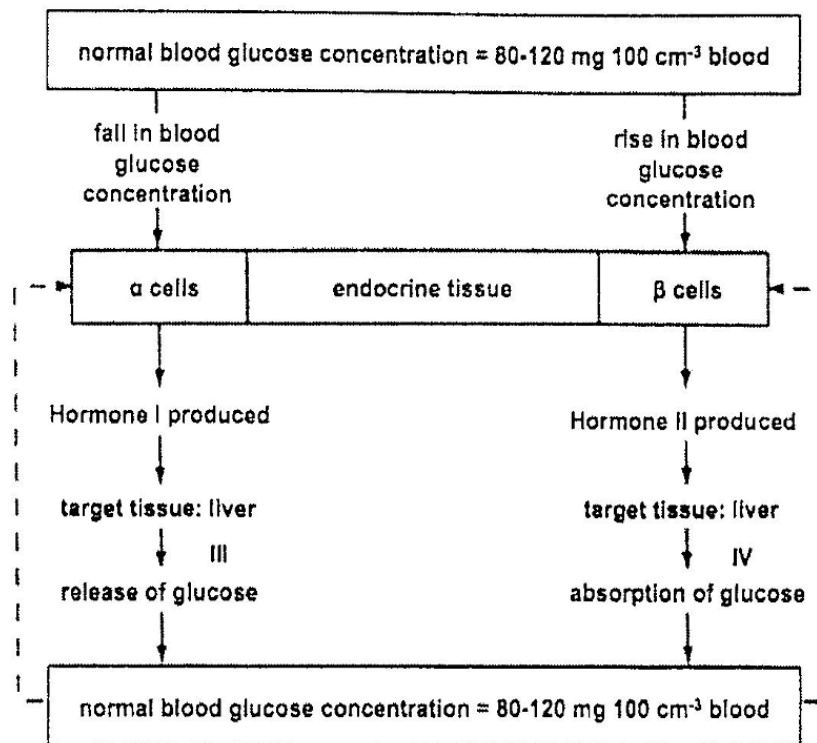
27 Some statements about the generation of an action potential in a neurone cell are listed below.

- I** A weak stimulus will result in an insignificant influx of Na^+ ions and the membrane returns to resting potential very quickly. No action potential is initiated.
- II** The generation of an action potential involves voltage-gated sodium ion channels, voltage-gated potassium ion channels, sodium potassium pumps.
- III** Another action potential can be generated in both the absolute and relative refractory period but will require a stronger than normal stimulus.
- IV** Without ATP, it is impossible to initiate an action potential as the membrane will not be polarized in the first place.

Which of the statements above is/are true?

- A** I only
- B** II and III
- C** I and IV
- D** I, II and IV

- 28 The diagram below shows the role of an endocrine tissue in controlling blood glucose concentration.

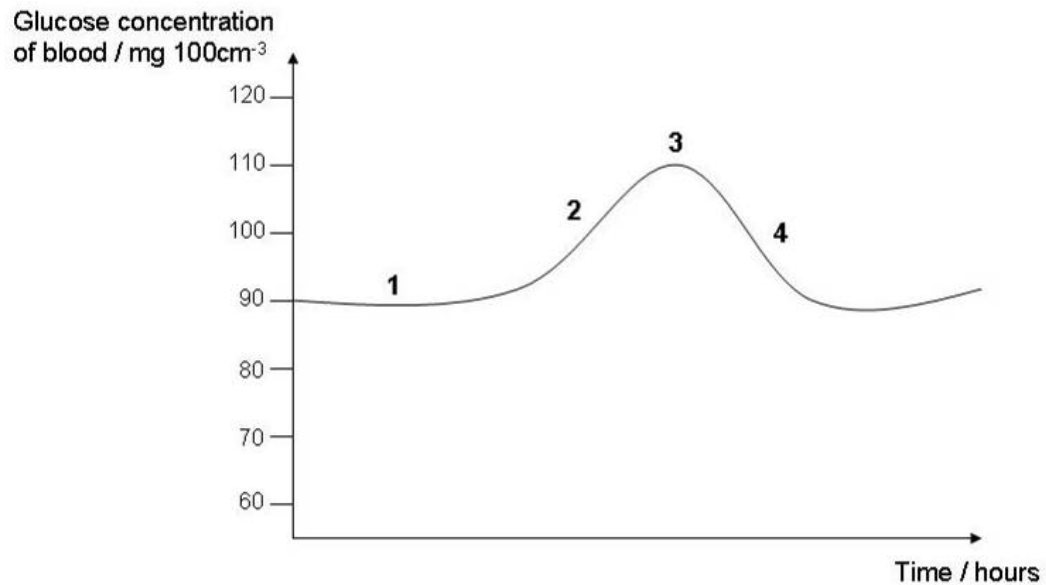


Which of the following statements is/are true?

- I Small amounts of hormone I and II inducing a large response in liver tissue demonstrates positive feedback
- II Hormones I and II inducing different responses from the same target tissue is due to the hormones binding to different receptors on the liver cell surface membrane.
- III The binding of Hormone I to receptors on liver cell surface membrane leads to the activation of second messenger cAMP

- A I only
- B II only
- C I and III
- D II and III

- 29 The graph below shows the changes of blood glucose level in a normal healthy person who consumes a meal.

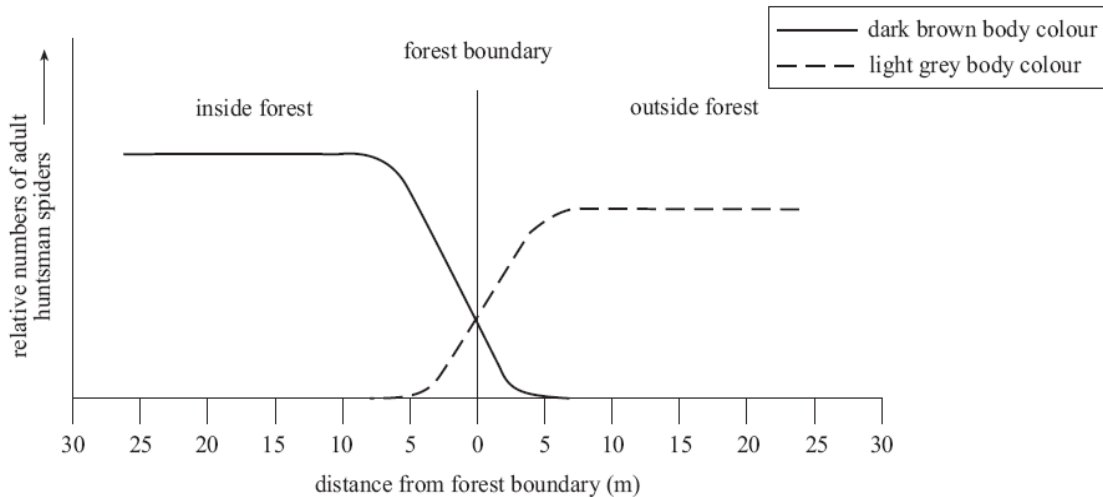


What is represented at each stage?

	negative feedback	set point	effector cells activated
A	4	1	3
B	4	1	2
C	3	1	4
D	3	1	2

- 30 Which feature does **not** support Darwin's theory of natural selection?
- A** adaptations to the environment, e.g. increased density of fur in bears in cold climates
 - B** homologous molecular structures, e.g. ATP in diverse organisms
 - C** similar anatomy, e.g. same number of neck vertebrae among mammals
 - D** similar structures for specific functions, e.g. fin of whale and shark

- 31 The graph below shows the distribution of huntsman spiders at a forest boundary:



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living inside the forest and others were found living just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

Which of the following can be **least** inferred from the graph?

- A The type of selective advantage inside the forest is different from the type of selective advantage outside the forest.
- B The plateau in the population number as seen inside the forest and outside the forest is due to competition among the adult spiders for limited resources.
- C The lower plateau of spider population outside the forest compared to that of inside the forest is due to the presence of additional selection pressure existing outside the forest.
- D Dark brown huntsman spiders are not eaten by birds inside the forest as their colour allows them to camouflage and hence provides a selective advantage.

32 The rate of divergence of different proteins is shown in the table below.

Protein	Amino acid substitutions per site per billion years
Fibrinopeptides	8.3
Lysozyme	2.0
α -globin	1.2
Insulin	0.44
Cytochrome <i>c</i>	0.3
Histone H4	0.01

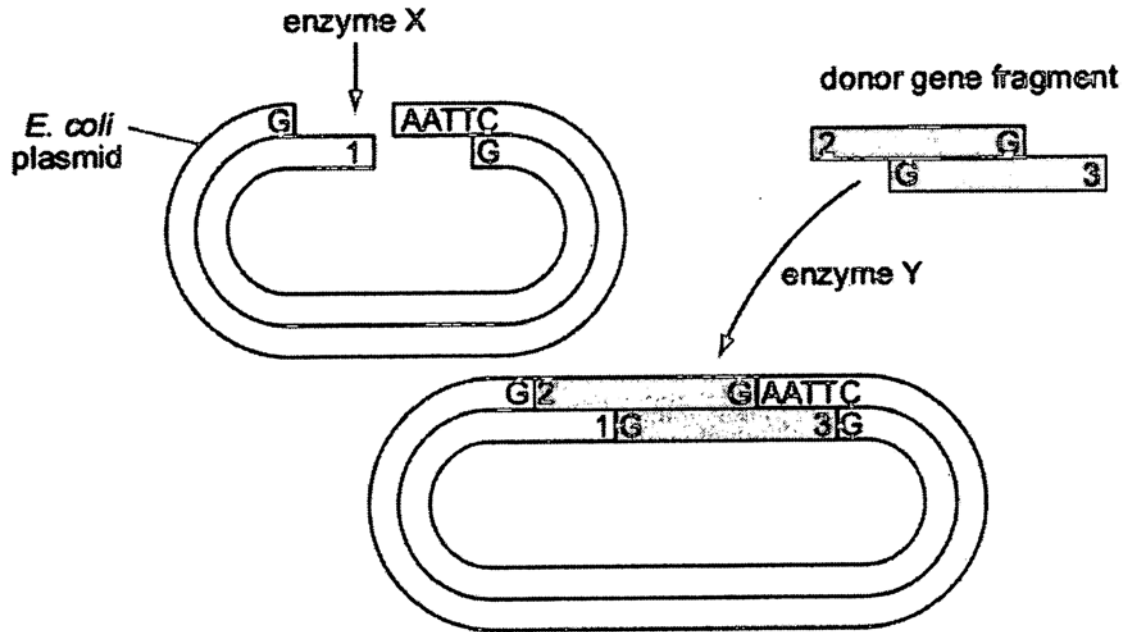
What does this data tell you about the neutral theory of molecular evolution?

- A** Genetic drift plays the greatest role in the gene for fibrinopeptides as the protein has the largest amino acid substitutions per site per billion years.
- B** Some essential proteins allow for fewer amino acid substitutions per site than others in order to maintain the same function.
- C** The rate of neutral mutations that result in the amino acid substitutions for all proteins is the same.
- D** All of these proteins are found in a common ancestor of all Earth's known living organisms before they started evolving at the molecular level.

33 A method used to detect a particular DNA sequence within a mixture of many DNA fragments is

- A** DNA sequencing
- B** DNA fingerprinting
- C** Colony hybridization
- D** Southern blotting

34 The diagram shows some stages in making recombinant DNA using *E. coli* plasmids.



Which are the correct enzymes X and Y and order of bases at 1, 2 and 3?

	X	Y	1	2	3
A	Ligase	Restriction enzyme	TTAA	AATT	TTAA
B	Restriction enzyme	Ligase	TTAA	AATT	TTAA
C	Ligase	Restriction enzyme	AATT	TTAA	AATT
D	Restriction enzyme	Ligase	AATT	TTAA	AATT

35 A plasmid has two antibiotic resistance genes, one for ampicillin and one for tetracycline. It is treated with a restriction enzyme that cuts in the middle of the ampicillin gene. DNA fragments containing a human globin gene were cut with the same enzyme. The plasmids and fragments are mixed, treated with ligase, and used to transform bacterial cells. Clones that have taken up the recombinant DNA are the ones that

- A are blue and can grow on agar plates with both antibiotics.
- B can grow on agar plates with ampicillin but not with tetracycline.
- C can grow on agar plates with tetracycline but not with ampicillin.
- D cannot grow on agar plates with any antibiotics.

- 36 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

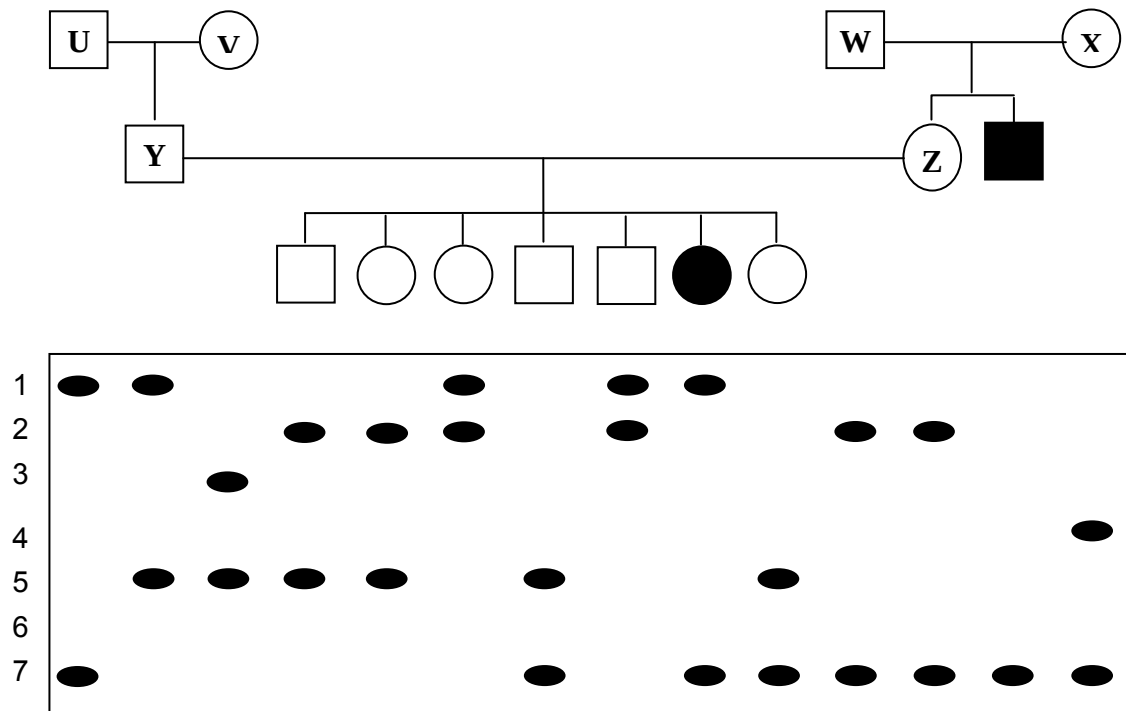
Template

5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'
3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A 5' GTCCAGC 3' & 5' CCTGAAC 3'
B 5' ATTCGGA 3' & 5' GATCTCC 3'
C 5' GGACTTG 3' & 5' GCTGGAC 3'
D 5' AUUCGGA 3' & 5' GAUCUCC 3'

- 37 A DNA polymorphism with 7 alleles,(1-7), is linked to the gene causing Disease H (an autosomal recessive disease).



In which individual has crossing over taken place between the DNA polymorphism and the gene loci?

- A U only
B V only
C W and X only
D Y and Z only

- 38** Which uses of the information from the human genome project are generally considered to be unethical?
- 1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases.
 - 2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships.
 - 3 cytologists developing tests for only some defective genes.
 - 4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes.
 - 5 genetic counselors giving specific lifestyle information only to people genetically predisposed to risks.
 - 6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes.
- A 1 and 3
B 1 and 6
C 2 and 5
D 3 and 4
- 39** Somaclonal variation may arise during *in vitro* culturing of plant tissues because
- A crossing over occurs between homologous chromosomes during meiosis in the cultured plant cells, leading to genetic recombination.
- B some cells are committed to differentiation into specialized plant tissues, therefore may silence the expression of certain genes.
- C use of large amount of growth regulators during culturing will lead to high growth rate of cells, increasing the rate of mutation in plant cells.
- D bacteriophage inserts DNA sequences containing genes from other plant cells into the cultured cells' genome, leading to genetic variation.
- 40** What are the normal functions of stem cells in a living human?

	Functions		
A	Differentiating into many kinds of cells in a 3-5 day old of human embryo	Producing insulin in pancreas damaged by type 1 diabetes	Cells that can differentiate into bone cells in the skeleton
B	Differentiating into many kinds of cells in a 3-5 day old of human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into cardiac muscle cells in the heart
C	Differentiating into only one kind of cells in a 3-5 day old of human embryo	Producing dopamine in the brains of people with Parkinson's	Cells that can differentiate into cartilage cells in the joints
D	Differentiating into only one kind of cells in a 3-5 day old of human embryo	Producing differentiated cells that can be used to screen new drugs	Cells that can differentiate into nerve cells in the brain